

Supporting Information

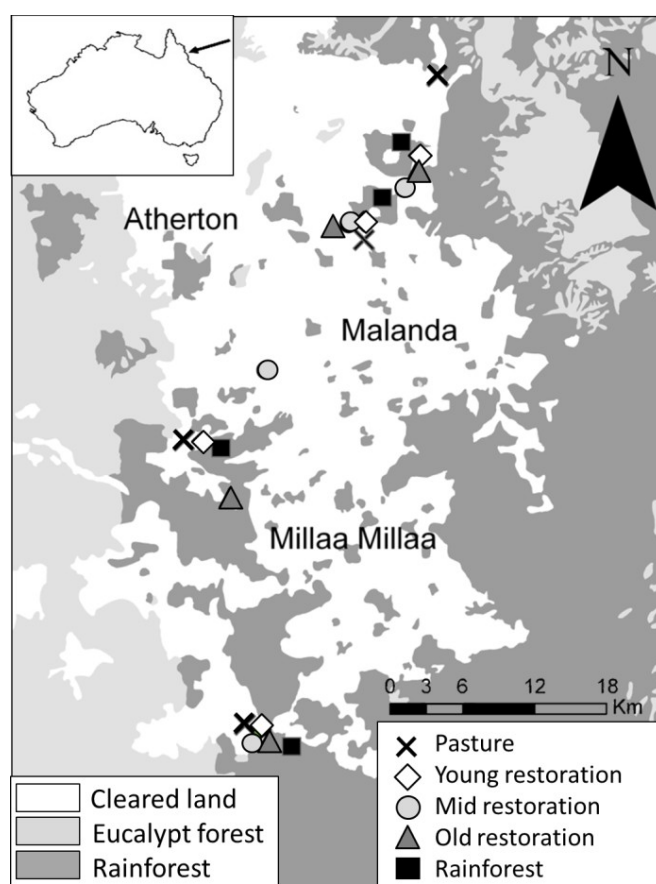


Figure S1 Map of the study area, showing the 20 study sites and areas of cleared forest, Eucalypt forest and rainforest. Each site comprised four sampling points (pitfall traps) for the measurement of community attributes and four experimental dung baits for the measurement of ecosystem functions.

Table S1 Abundance of each species and their assigned functional trait values, including the habitat category within which the majority of individuals were recorded ('habitat preference'): P= pasture; YR= young restoration; MR= mid-age restoration; OR= old restoration; RF = rainforest.

Species	Taxonomic authority	Abundance	Body mass (mg)	Behavioural guild	Diel activity	Diet preference	Diet breadth	Habitat preference
<i>Amphistomus complanatus</i>	Matthews (1974)	333	9.60	Rollers	Nocturnal	Dung	2	YR; MR; OR; RF
<i>Amphistomus NQ3</i>	NA	235	18.31	Rollers	Nocturnal	Dung	2	RF
<i>Amphistomus NQ4</i>	NA	18	3.21	Rollers	NA	Dung	2	OR; RF
<i>Amphistomus NQ5</i>	NA	302	1.55	Rollers	Nocturnal	Dung	2	OR; RF
<i>Amphistomus pygmaeus</i>	Matthews (1974)	7	3.30	Rollers	Nocturnal	Dung	1	RF
<i>Aptenocantho winyar</i>	Storey & Monteith (2000)	4	9.00	Rollers	NA	Dung	1	RF
<i>Boletoscapter cornutus</i>	Matthews (1974)	68	8.50	Rollers	Nocturnal	Mushroom	1	MR; OR; RF
<i>Coptodactyla depressa</i>	Matthews (1976)	409	51.25	Tunnelers	Nocturnal	Both	2	MR; OR; RF
<i>Coptodactyla onitoides</i>	Matthews (1976)	79	76.80	Tunnelers	Nocturnal	Both	2	MR; OR
<i>Demarziella interrupta</i>	Matthews (1976)	27	2.95	Tunnelers	Nocturnal	Dung	2	P
<i>Lepanus globulus</i>	Matthews (1974)	11	4.43	Rollers	NA	Both	2	RF
<i>Lepanus latheticus</i>	Matthews (1974)	10	0.70	Rollers	NA	Mushroom	2	RF
<i>Lepanus nitidus (large)</i>	Matthews (1974)	13	3.50	Rollers	Diurnal	Both	2	RF
<i>Lepanus nitidus (small)</i>	Matthews (1974)	115	1.40	Rollers	Diurnal	Mushroom	2	RF
<i>Lepanus NQ11</i>	NA	1	0.70	Rollers	NA	NA	NA	OR
<i>Lepanus NQ3</i>	NA	3	0.75	Rollers	NA	Mushroom	2	RF
<i>Lepanus NQ5</i>	NA	1	0.87	Rollers	NA	Mushroom	NA	RF
<i>Lepanus palumensis</i>	Matthews (1974)	1	0.80	Rollers	NA	Mushroom	1	RF
<i>Lepanus villosus</i>	Matthews (1974)	12	0.65	Rollers	Diurnal	Mushroom	1	RF
<i>Onthophagus bornemisszanus</i>	Matthews (1972)	2	20.00	Tunnelers	NA	NA	NA	P
<i>Onthophagus bundara</i>	Storey & Weir (1990)	1	1.29	Tunnelers	NA	NA	NA	RF
<i>Onthophagus capelliformis</i>	Matthews (1972)	113	25.47	Tunnelers	Nocturnal	Dung	1	YR; MR; OR; RF

<i>Onthophagus capella</i>	Matthews (1972)	17	52.95	Tunnelers	Nocturnal	Dung	1	P; YR
<i>Onthophagus cuniculus</i>	Matthews (1972)	31	19.70	Tunnelers	Diurnal	Mushroom	2	P; YR
<i>Onthophagus darlingtoni</i>	Matthews (1972)	3	15.62	Tunnelers	Nocturnal	Dung	2	RF
<i>Onthophagus dicranocerus</i>	Matthews (1972)	18	31.04	Tunnelers	Nocturnal	Dung	1	MR; OR; RF
<i>Onthophagus gulmarri</i>	Matthews (1972)	4	9.00	Tunnelers	Nocturnal	Dung	1	MR; RF
<i>Onthophagus millamilla</i>	Matthews (1972)	59	4.38	Tunnelers	Diurnal	Dung	2	MR; OR; RF
<i>Onthophagus mundill</i>	Matthews (1972)	1	61.00	Tunnelers	Nocturnal	NA	NA	RF
<i>Onthophagus nigriventris</i>	D'Orbigny (1902)	2	38.31	Tunnelers	Diurnal	NA	NA	OR
<i>Onthophagus paluma</i>	Matthews (1972)	1	28.50	Tunnelers	Diurnal	Dung	1	P
<i>Onthophagus rubicundulus</i>	Matthews (1972)	15	1.86	Tunnelers	Diurnal	Dung	2	RF
<i>Onthophagus semimetallicus</i>	Matthews (1972)	12	12.0	Tunnelers	Nocturnal	Dung	1	MR
<i>Onthophagus wagamen</i>	Matthews (1972)	62	5.70	Tunnelers	Diurnal	Dung	1	RF
<i>Onthophagus waminda</i>	Matthews (1972)	84	1.93	Tunnelers	Diurnal	Dung	1	YR; MR; OR; RF
<i>Onthophagus yungaburra</i>	Matthews (1972)	136	2.57	Tunnelers	Diurnal	Dung	1	RF
<i>Temnoplectron aeneopiceum</i>	Matthews (1974)	1	4.56	Rollers	Nocturnal	Unknown	1	RF
<i>Temnoplectron bornemisszai</i>	Matthews (1974)	18	63.50	Rollers	Nocturnal	Dung	2	RF
<i>Temnoplectron politulum</i>	Matthews (1974)	387	18.56	Rollers	Nocturnal	Both	2	OR; RF

Appendix S1 Functional trait assignment methods

Where information on a species was unavailable, NAs were used. This was necessary for eight species for diet preference and breadth, and for ten species for diel activity and behavioural guild. In order to calculate functional diversity metrics, traits were given equal weighting and species were weighted by their relative abundance. Sample sites for which there were less than three species recorded ($n=4$) were excluded from any analysis involving the functional diversity metrics, because functional diversity indices cannot be calculated from less than three species.

Behavioural guild, diel activity and body mass

Information on species' behavioural guilds and diel activity was obtained from the literature (Matthews 1972, 1974, 1976, Storey and Weir 1989, Storey and Monteith 2000) and from observations conducted by the authors (MD, GM and RM). Dung beetles were classified according to their resource relocation behaviour into either tunnelers, which bury dung directly beneath the dung pile; and rollers, which transport and bury dung some distance away from the dung pile (Hanski and Cambefort 1991). To calculate body mass, 1-20 specimens (unknown sex) of each species were dried in a forced draught oven at 60°C to a constant weight, and weighed using a 0.0001 g precision balance.

Diet preference and breadth

Diet preference was investigated using traps alternately baited with wallaby dung ($n = 80$ traps) and mushrooms ($n = 80$ traps). Trap design was identical for traps baited with both bait types. We used the proportion of individuals of each species attracted to a certain bait to determine bait specificity. Species of which >80% individuals were recorded in traps baited with dung were categorised as 'dung specialists'; species of which >80% individuals were recorded in traps baited with mushroom were categorised as 'mushroom specialists'; and species of which 25-75% individuals were recorded in traps baited with dung were categorised as 'diet generalists'. Diet breadth was calculated as the number of bait types that a species was attracted to (species caught in both dung and mushroom baited traps = 2; species caught only in either dung or mushroom = 1). The minimum number of individuals of each species required to calculate diet preference and breadth was $n = 3$. Information on diet preference and breadth has important implications for resource partitioning between functional groups.

Table S2 Structure of generalised linear mixed models for determining the effects of restoration age and habitat category on species, functional diversity and community metrics and ecological functions.

Response variable	Fixed effect: Restoration age			Fixed effect: Habitat category		
	Random effect(s)	Error distribution	Transformation	Random effect(s)	Error distribution	Transformation
Species metrics						
Species richness	Block	Poisson		Block	Poisson	
Abundance	Block	Negative binomial		Block	Negative binomial	
Biomass	Block	Gamma (log link)		Block	Gaussian	log ₁₀
Species diversity	Block	Gaussian		Block	Gaussian	
Rarefied species richness	Block	Gaussian	sqrt	Block	Gaussian	sqrt
Estimated species richness	Block	Negative binomial		Block	Gaussian	sqrt
Species evenness	Block	Gaussian		Block	Gaussian	
Functional diversity metrics						
Functional richness	Block	Gaussian		Block	Gaussian	
Functional evenness	Block	Gaussian		Block	Gaussian	
Functional divergence	Block	Gaussian		Block	Gaussian	
Functional dispersion	Block	Gaussian		Block	Gaussian	
Community metrics						
Bray-Curtis similarity to Pasture/ Rainforest	Block	Beta		Block	Beta	
Ecological functions						
Dung removal	Block	Gaussian	log ₁₀	Block	Gaussian	log ₁₀
Seed dispersal	Block	Beta		Block	Beta	
Soil excavation	Block	Gamma (log link)		Block	Gamma (log link)	
Multifunctionality	Block	Gaussian		Block	Gaussian	

Table S3 Structure of global models for determining the effects of species richness (SpRic), abundance (Abun), biomass (Biom), functional richness (FRic), functional evenness (FEve), functional dispersion (FDis) and functional divergence (FDiv) on dung removal (Dung), seeds dispersal (Seeds), soil excavation (Soil) and multifunctionality (Multi).

Global model	Model	Random effect(s)
$\log(\text{Dung}) \sim \text{SpRic} + \sqrt{\text{Abun}} + \sqrt{\text{Biom}} + \text{FRic} + \text{FEve} + \text{FDis} + \text{FDiv}$	lmm	Block
$\log(\text{Seeds}) \sim \text{SpRic} + \sqrt{\text{Abun}} + \sqrt{\text{Biom}} + \text{FRic} + \text{FEve} + \text{FDis} + \text{FDiv}$	lmm	Block
$\log(\text{Soil}) \sim \text{SpRic} + \sqrt{\text{Abun}} + \sqrt{\text{Biom}} + \text{FRic} + \text{FEve} + \text{FDis} + \text{FDiv}$	lmm	Block
$\text{Multi} \sim \text{SpRic} + \text{Abun} + \text{Biom} + \text{FRic} + \text{FEve} + \text{FDis} + \text{FDiv}$	lmm	Block

*Number of individuals (abundance) and biomass were square root transformed to satisfy assumptions of normality

Table S4 Summary of total abundance, observed (Sobs) and estimated (Sest) species richness, and proportion of species detected (Sobs / Sest) in each habitat category. Superscripts represent pairwise differences at the $P \leq 0.05$ level.

Measure	Pasture	Young restoration	Mid restoration	Old restoration	Rainforest
Abundance	78 ^a	138 ^a	537 ^b	957 ^{bc}	1607 ^c
Sobs	7 ^a	11 ^a	20 ^b	22 ^b	32 ^c
Sest	8.97 ^a	13.56 ^a	28.20 ^b	32.48 ^b	37.70 ^b
Sobs/ Sest	0.78 ^a	0.81 ^a	0.71 ^a	0.68 ^a	0.85 ^a

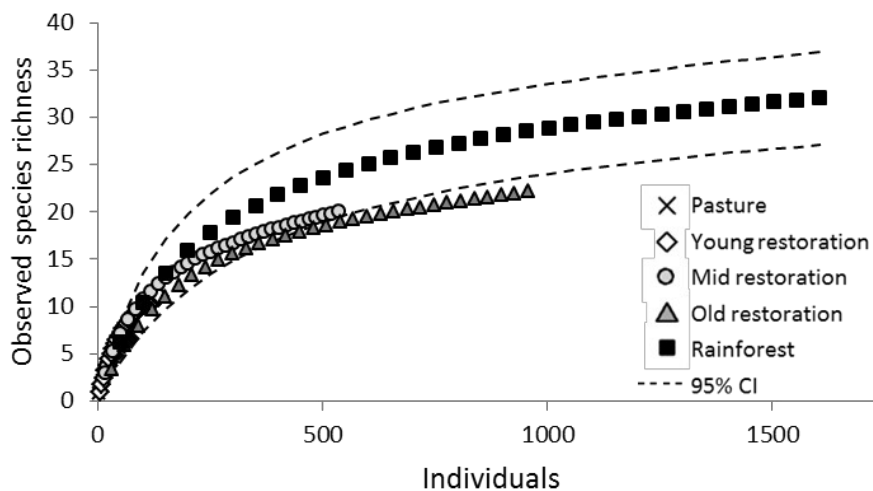


Figure S2 Species accumulation curves constructed using sample-based rarefaction curves for pasture, rainforest and the three restoration categories (scaled to show the number of individuals). Dashed line represents 95% confidence interval (CI) of rainforest.

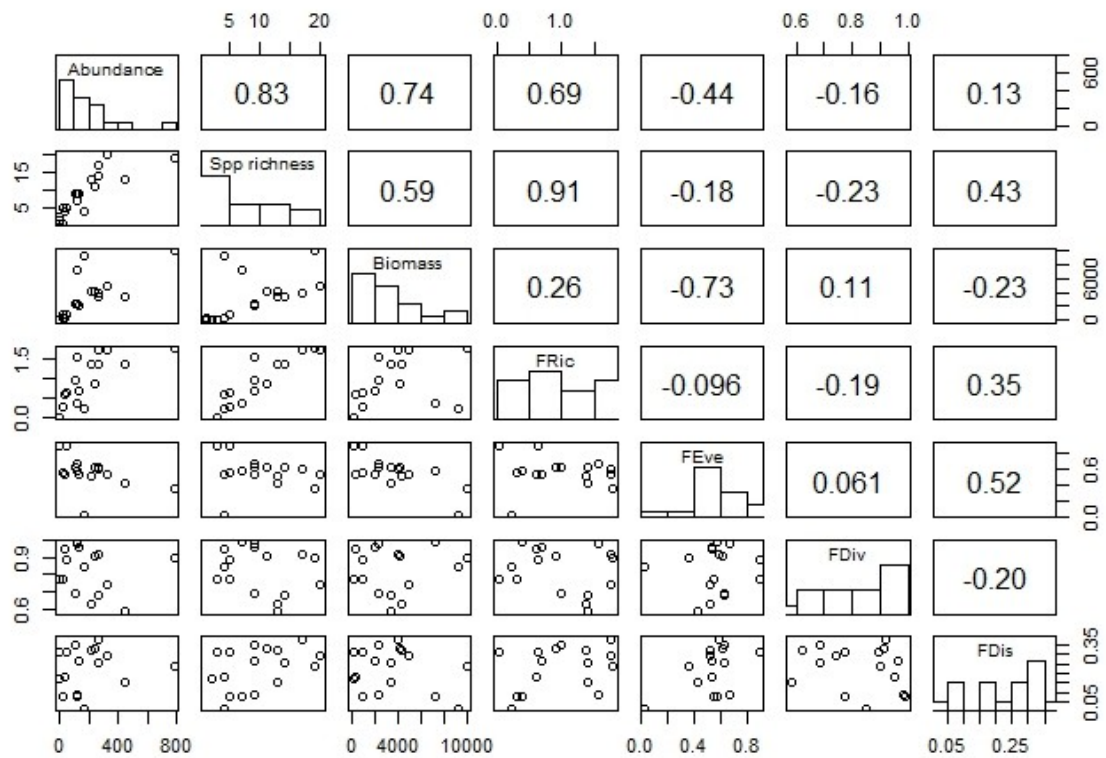


Figure S3 Associations between dung beetle community attributes: bivariate plots (lower panels), distributions (diagonal), and Pearson's ρ (upper panels).

Table S5 Pairwise comparisons of Bray Curtis assemblage similarity between different habitat categories. P = pasture; YR = young restoration; MR = mid-age restoration; OR = old restoration; RF = rainforest. Asterisks indicate significant differences between categories at the 0.05 level. Degrees of freedom in all models = 1.

	P		YR		MR		OR	
	R ²	Pr(>F)	R ²	Pr(>F)	R ²	Pr(>F)	R ²	Pr(>F)
YR	0.1							
	4	0.472	-	-	-	-	-	-
MR	0.4	0.033	0.2					
	1	*	3	0.076	-	-	-	-
OR	0.3	0.029	0.1		0.0			
	6	*	8	0.266	5	0.824	-	-
RF	0.5	0.023	0.2	0.025	0.2		0.1	
	1	*	6	*	1	0.250	4	0.502

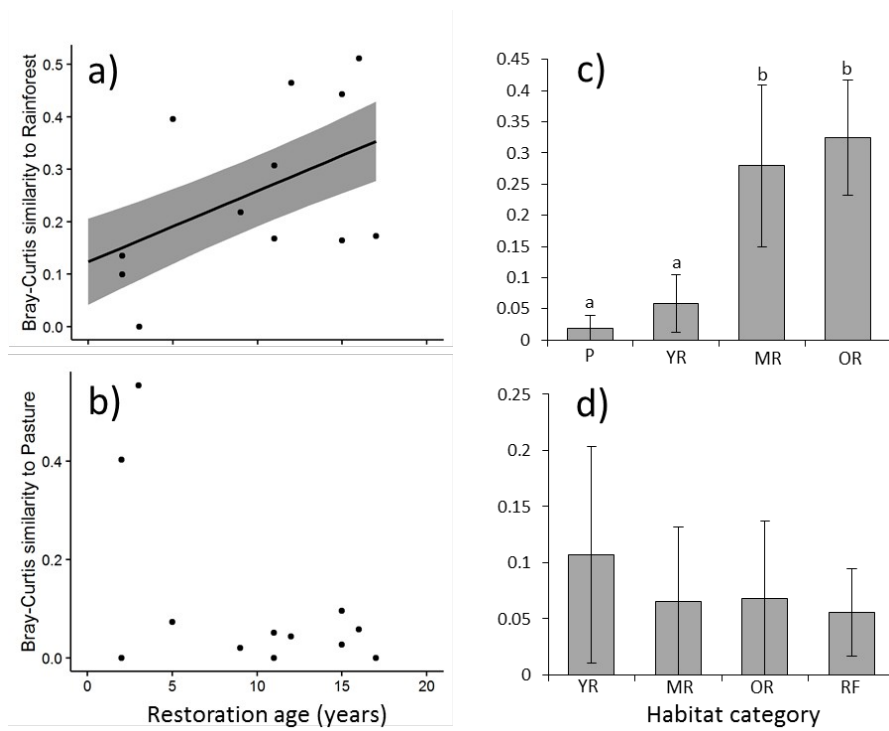


Figure S4 Relationship between restoration age and dung beetle assemblage similarity (Bray–Curtis index) to primary forest and pasture (a-b). Mean $\pm$ SE dung beetle assemblage similarity (Bray–Curtis index) to primary forest and pasture in the different habitat categories (c-d). P = pasture; YR = young restoration; MR = mid-age restoration; OR = old restoration; RF = rainforest. Unlike letters indicate significant differences ($P < 0.05$).

Table S6 Support for generalized linear mixed models predicting multifunctionality, dung removal, seed dispersal and soil excavation in relation to dung beetle community attributes and functional diversity metrics.

Predictor	log(L)	ΔAIC_c	w_i
Multifunctionality			
FDiv	-17.04	0.00	0.309
FDis	-17.08	0.08	0.297
FEve	-17.37	0.66	0.223
Biomass	-18.02	1.95	0.116
FRic	-19.05	4.02	0.041
Abundance	-20.51	6.93	0.01
Species richness	-21.34	8.59	0.004
Dung removal			
FDis	-12.54	0.00	0.568
FEve	-13.83	2.57	0.157
FDiv	-13.86	2.65	0.151
Biomass	-14.78	4.49	0.060
FRic	-14.95	4.82	0.051
Abundance	-16.90	8.71	0.007
Species richness	-17.06	9.04	0.006
Seed dispersal			
FDis	-12.59	0.00	0.401
FDiv	-13.13	1.07	0.235
FEve	-13.41	1.64	0.176
Biomass	-13.85	2.52	0.114
FRic	-14.63	4.07	0.053
Abundance	-16.02	6.86	0.013
Species richness	-16.46	7.73	0.008
Soil excavation			
FDis	-16.23	0.00	0.432
FDiv	-16.56	0.66	0.310
FEve	-17.40	2.34	0.134
FRic	-17.84	3.21	0.087
Biomass	-18.83	5.19	0.032
Abundance	-20.97	9.49	0.004
Species richness	-21.68	10.90	0.002

* Abundance and Biomass were square root transformed to satisfy assumptions of normality. Block was included as a random effect in each model (n = 20 sites). There were 4 parameters in each model and 5 degrees of freedom. log(L) is the log likelihood; ΔAIC_c is the difference between the AIC_c of each model and that of the top model; and w_i is the Akaike weight.

Literature cited

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